Thermodynamics

01.03 · Original practice problems

Show your reasoning, use SI units where appropriate, and state any assumptions. Constants are supplied in the questions that need them.

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A MAIS Physics & Astronomy Club study resource. These are original exercises, not official IOAA problems.

PROBLEMS

01 An ideal-gas cloud

A uniform gas cloud contains 2.0×10^{50} particles in a volume of 1.0×10^{45} m^3 at 100 K. Treat it as an ideal gas and use $k_B = 1.38 \times 10^{-23}$ J K^{-1}. (a) Find the number density and pressure. (b) If the gas is heated at fixed volume to 300 K, find the new pressure. (c) State a condition under which the ideal-gas approximation might fail.

02 Heating and work

A monatomic ideal gas contains 2.0 mol. It expands at constant pressure 1.0×10^5 Pa from 0.020 m^3 to 0.030 m^3. Use $R = 8.31$ J mol^{-1} K^{-1} and $U = 3nRT/2$. (a) Find the initial and final temperatures. (b) Find the work done by the gas and the change in its internal energy. (c) Use the first law to find the heat added to the gas.

03 Planetary equilibrium

A planet absorbs a fraction 0.70 of the incident stellar flux $F = 1360$ W m^{-2}. Assume a spherical planet, uniform temperature, no internal heat and blackbody emission over its entire surface. Use $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}. (a) Explain the factor of four relating absorbed flux averaged over the whole surface to F. (b) Find the equilibrium temperature. (c) Give one reason the actual surface temperature could differ.

04 Energy transport

Consider a hot stellar interior and a cooler surface. (a) Explain the difference between radiative and convective energy transport. (b) Why must there be a temperature gradient for a net outward radiative energy flow? (c) A gas parcel rises and remains warmer than its surroundings at the same pressure. Explain qualitatively why it may continue to rise.